MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. Rate of x^2 between 1 and 2:

   A. 2
   B. 3
   C. 1
   D. 4

2. If $f(1) = 4$, $f'(1) = 2$, tangent:

   A. $y = 2x + 2$
   B. $y = 4x + 2$
   C. $y = 2x + 4$
   D. $y = x + 3$

3. The derivative at this point is

   A. a function
   B. an area
   C. an x-coordinate
   D. a real number if the derivative exists

4. For x^2 at 3, derivative:

   A. 9
   B. 3
   C. 6
   D. 12

5. The approximation of $f(a + h)$ is

   A. $f(a)f'(a)$
   B. $f(a) + hf'(a)$
   C. $f(a) + f'(h)$
   D. $hf(a)$

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. $\sqrt{x}$ at 0 is

   A. differentiable with derivative 0
   B. differentiable with derivative 1
   C. not differentiable
   D. undefined

2. $\dfrac{|h|}{h}$ for $h < 0$ equals

   A. -1
   B. 1

C. 0

D. h

3. A tangent with slope 0 is

A. vertical

B. oblique

C. impossible

D. horizontal

4. Position in m and time in s: unit of the derivative

A. m^2

B. $\mathrm{m\,s}^{-1}$

C. $\mathrm{s\,m}^{-1}$

D. m

5. For f differentiable at a, replace $f(a+h)$ by $f(a)+hf\prime(a)$ is in general

A. an exact identity

B. a definition of h

C. a local approximation

D. always false, even for an affine function

True or false? Justify.

1. A tangent can never meet a curve again.

2. The absolute value function is differentiable at 0.

3. An affine function $mx+p$ has derivative m at every real number.

Personal review

I give the domain and h non-zero; I read a slope; I use the point of tangency; I distinguish exact and approximate.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.